Stage 1 Hard Drill — With Solutions

Chapter 5.3 — Geometric Probability & 2D Integrals

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the picture is drawn. Master rule: *draw the region, shade the event, measure the area.*

Q1 — Closer to the Centre Than to Any Side

A point is uniform in the unit square. Find the probability that it is closer to the *centre* than to the *nearest side*.

Solution. Centre the square: $u = x - \frac{1}{2}$, $v = y - \frac{1}{2}$, so $|u|, |v| \le \frac{1}{2}$. The distance to the nearest side is $\frac{1}{2} - \max(|u|, |v|)$. By the 8-fold symmetry, work in the piece $0 \le v \le u \le \frac{1}{2}$ (area $\frac{1}{8}$), where the nearest side is at distance $\frac{1}{2} - u$:

$$\sqrt{u^2 + v^2} < \tfrac{1}{2} - u \iff v^2 < \tfrac{1}{4} - u \iff v < \sqrt{\tfrac{1}{4} - u}$$

Together with $v \le u$. The two curves meet where $u^2 = \frac{1}{4} - u$, i.e. $u_0 = \frac{\sqrt{2}-1}{2}$ (note $\frac{1}{4} - u_0 = u_0^2$). Then

$$\text{Area} = \int_0^{u_0} u\,du + \int_{u_0}^{1/4} \sqrt{\tfrac{1}{4} - u}\,du = \tfrac{u_0^2}{2} + \tfrac{2}{3}u_0^3$$

Multiply by 8 and use $u_0^2 = \frac{3-2\sqrt{2}}{4}$, $u_0^3 = \frac{5\sqrt{2}-7}{8}$:

$$\mathbb{P} = 4u_0^2 + \tfrac{16}{3}u_0^3 = (3 - 2\sqrt{2}) + \tfrac{10\sqrt{2}-14}{3} = \boxed{\tfrac{4\sqrt{2}-5}{3} \approx 0.219}$$

⋆ *Intuition:* "equidistant from a point and a line" is a *parabola* (focus–directrix). The event is the region bounded by four parabolas meeting near the centre — a curved boundary is the cue to integrate rather than count triangles.

Q2 — Uniform Under a Parabola

(X, Y) is uniform on $\{0 < y < x^2,\ 0 < x < 1\}$. Find $\text{Cov}(X, Y)$ and $\mathbb{P}(Y > X^3)$.

Solution. Area $= \int_0^1 x^2 dx = \frac{1}{3}$, so the density is 3.

$$f_X(x) = \int_0^{x^2} 3\,dy = 3x^2 \ \Rightarrow\ \mathbb{E}[X] = \int_0^1 3x^3 dx = \tfrac{3}{4}$$

$$\mathbb{E}[Y] = \int_0^1 \int_0^{x^2} 3y\,dy\,dx = \int_0^1 \tfrac{3}{2}x^4 dx = \tfrac{3}{10}$$

$$\mathbb{E}[XY] = \int_0^1 x \cdot \tfrac{3}{2}x^4 dx = \tfrac{1}{4}$$

$$\text{Cov}(X, Y) = \tfrac{1}{4} - \tfrac{3}{4} \cdot \tfrac{3}{10} = \boxed{\tfrac{1}{40}}$$

For $\{Y > X^3\}$: since $x^3 < x^2$ on $(0, 1)$, it is the sliver $x^3 < y < x^2$:

$$\mathbb{P} = 3\int_0^1 (x^2 - x^3)\,dx = 3\left(\tfrac{1}{3} - \tfrac{1}{4}\right) = \boxed{\tfrac{1}{4}}$$

▶ *Speed trick:* the density is 1/Area, so *compute the area first.* Every expectation then reduces to one iterated integral with the inner limits read off the region.

Q3 — Between a Line and a Hyperbola

X, Y are iid Uniform$(0, 1)$. Find $\mathbb{P}\big(X + Y \le 1 \text{ and } XY \ge \frac{1}{8}\big)$.

Solution. The line $y = 1 - x$ and the hyperbola $y = \frac{1}{8x}$ meet where $x(1 - x) = \frac{1}{8}$:

$$8x^2 - 8x + 1 = 0 \ \Rightarrow\ x_{1,2} = \tfrac{2 \mp \sqrt{2}}{4} \quad (x_1 + x_2 = 1,\ x_2 - x_1 = \tfrac{\sqrt{2}}{2})$$

Between the roots the hyperbola lies below the line, so

$$\text{Area} = \int_{x_1}^{x_2} \Big(1 - x - \frac{1}{8x}\Big)dx$$

First piece: $\int (1 - x)dx = (x_2 - x_1)\big[1 - \frac{x_1 + x_2}{2}\big] = \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{4}$.
Second piece: $\frac{1}{8}\ln\frac{x_2}{x_1} = \frac{1}{8}\ln\frac{2+\sqrt{2}}{2-\sqrt{2}} = \frac{1}{8}\ln(1+\sqrt{2})^2 = \frac{1}{4}\ln(1+\sqrt{2})$.

$$\mathbb{P} = \boxed{\tfrac{1}{4}\Big(\sqrt{2} - \ln(1 + \sqrt{2})\Big) \approx 0.1332}$$

▶ *Speed trick:* find the crossing points *first* — they are the integration limits — and check which curve is on top between them (a quick test point settles it).

Q4 — The Meeting Problem With Unequal Patience

Two traders arrive independently and uniformly in a 60-minute window. Trader A waits 10 minutes, trader B waits 20 minutes, then each leaves. Find $\mathbb{P}$(they meet).

Solution. Let A arrive at x and B at y. Their presence intervals $[x, x + 10]$ and $[y, y + 20]$ overlap iff

$$y \le x + 10 \quad \text{and} \quad x \le y + 20$$

The two *miss* regions in the 60×60 square are corner triangles:

$$y > x + 10:\ \text{legs } 50,\ \text{area } \tfrac{50^2}{2} = 1250$$

$$x > y + 20:\ \text{legs } 40,\ \text{area } \tfrac{40^2}{2} = 800$$

$$\mathbb{P}(\text{meet}) = 1 - \frac{1250 + 800}{3600} = \frac{1550}{3600} = \boxed{\tfrac{31}{72} \approx 0.431}$$

▶ *Speed trick:* $\mathbb{P}(\text{miss}) = \dfrac{(T - a)^2 + (T - b)^2}{2T^2}$ for waits a, b in a window of length T. Equal waits $a = b$ recover $1 - (1 - a/T)^2$.

Q5 — The Ratio of Two Uniforms

X, Y are iid Uniform$(0, 1)$ and $R = X/Y$. Find (a) $\mathbb{P}(\frac{1}{2} \le R \le 2)$ and (b) $\mathbb{P}(\lfloor R \rfloor \text{ is odd})$.

Solution. $\{R \le t\} = \{x \le ty\}$ is a region cut by a line through the origin.
For $t \le 1$: a triangle with base 1 and height t, so $F(t) = \frac{t}{2}$.
For $t > 1$: the complement $\{x > ty\}$ is a triangle with legs 1 and $\frac{1}{t}$, so $F(t) = 1 - \frac{1}{2t}$.

(a) $F(2) - F(\frac{1}{2}) = \frac{3}{4} - \frac{1}{4} = \boxed{\frac{1}{2}}$.

(b) For integer $k \ge 1$:

$$\mathbb{P}(k \le R < k + 1) = \Big(1 - \tfrac{1}{2(k+1)}\Big) - \Big(1 - \tfrac{1}{2k}\Big) = \tfrac{1}{2}\Big(\tfrac{1}{k} - \tfrac{1}{k+1}\Big)$$

Sum over odd k (the value $\lfloor R \rfloor = 0$ is even and excluded):

$$\tfrac{1}{2}\Big(1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots\Big) = \boxed{\tfrac{\ln 2}{2} \approx 0.347}$$

⋆ *Intuition:* a ratio is a *slope*, and fixing the slope picks a wedge through the origin. Heavy tail: $\mathbb{P}(R > t) = \frac{1}{2t}$ decays only like $1/t$, which is why $\mathbb{E}[R] = \infty$.

Q6 — The Piece Containing the Midpoint (Inspection Paradox)

A unit stick is broken at two independent uniform points. Find the expected length of the piece that contains the midpoint $x = \frac{1}{2}$.

Solution. Let L be the left end of that piece (the nearest cut left of $\frac{1}{2}$, or 0) and R its right end. For $0 \le l < \frac{1}{2}$, $L > l$ iff at least one cut lands in $(l, \frac{1}{2})$, an interval of length $\frac{1}{2} - l$:

$$\mathbb{P}(L > l) = 1 - \big(\tfrac{1}{2} + l\big)^2$$

$$\mathbb{E}[L] = \int_0^{1/2} \Big[1 - \big(\tfrac{1}{2} + l\big)^2\Big] dl = \tfrac{1}{2} - \tfrac{1 - 1/8}{3} = \tfrac{5}{24}$$

By symmetry $\mathbb{E}[1 - R] = \frac{5}{24}$, so

$$\mathbb{E}[R - L] = 1 - 2 \cdot \tfrac{5}{24} = \boxed{\tfrac{7}{12} \approx 0.583}$$

⋆ *Intuition:* a typical piece has length $\frac{1}{3}$, yet the piece you *land in* averages $\frac{7}{12}$. Big pieces cover more of the stick, so a fixed point

lands in them more often (size-biased sampling). Cross-check: a *uniformly random* point lands in a piece of expected length $\mathbb{E}[a^2+b^2+c^2] = 3\cdot\frac{1}{6} = \frac{1}{2}$ — between the edge value $\frac{1}{3}$ and the midpoint value $\frac{7}{12}$, as it must be.

Q7 — Breaking Into Four: Quadrilaterals

A unit stick is broken at $n = 3$ independent uniform points into 4 pieces. Find (a) $\mathbb{P}$(the pieces form a quadrilateral), (b) the general formula for n cuts, and (c) $\mathbb{E}$[shortest piece].

Solution. A polygon exists iff the longest piece is shorter than the sum of the others, i.e. iff *every piece is* $< \frac{1}{2}$.

A given piece exceeds $\frac{1}{2}$ iff all n cuts fall on one side of it. By the same cut-counting argument as the 3-piece case,

$$\mathbb{P}(\text{piece} > t) = (1-t)^n \;\Rightarrow\; \mathbb{P}(\text{piece} > \tfrac{1}{2}) = 2^{-n}$$

At most one piece can exceed $\frac{1}{2}$, so these events are disjoint:

$$\mathbb{P}(\text{polygon}) = 1 - \frac{n+1}{2^n}$$

For $n=3$: $1-\frac{4}{8} = \boxed{\frac{1}{2}}$. (Check $n=2$: $1-\frac{3}{4} = \frac{1}{4}$.)

(c) Reserving t for each of the 4 pieces leaves a free stick of length $1-4t$:

$$\mathbb{P}(\min > t) = (1-4t)^3 \;\Rightarrow\; \mathbb{E}[\min] = \int_0^{1/4} (1-4t)^3\,dt = \boxed{\frac{1}{16}}$$

In general $\mathbb{E}[\min] = \frac{1}{(n+1)^2}$.

▶ *Speed trick:* "no bully" ⇒ polygon, and the bully events are disjoint, so the answer is always 1 − (pieces) × (one bully).

Q8 — Three Points on a Circle

Three points are chosen independently and uniformly on a circle. Find (a) $\mathbb{P}$(the triangle contains the centre), and (b) the expected length of the chord joining two such points on the unit circle.

Solution. (a) Fix one point A and cut the circumference there into a "stick" of length 2π; the other two points are two uniform cuts, and the three arcs between consecutive points are the pieces.

The centre lies inside the triangle iff *no arc is* $\geq \pi$ (an arc $\geq \pi$ means all three points sit in a half-circle). That is exactly the "every piece $< \frac{1}{2}$" event of the stick problem:

$$\mathbb{P} = \boxed{\frac{1}{4}}$$

(b) For two points at angular separation $\Delta \sim \text{Uniform}(0, 2\pi)$ the chord is $2\sin(\Delta/2)$:

$$\mathbb{E}[\text{chord}] = \frac{1}{2\pi}\int_0^{2\pi} 2\sin\tfrac{\Delta}{2}\,d\Delta = \frac{1}{2\pi}\cdot 2\cdot 4 = \boxed{\frac{4}{\pi} \approx 1.273}$$

⋆ *Intuition:* a circle is a stick with its ends glued, so "cut at one point" converts any circle problem into a stick problem — the same $\frac{1}{4}$ reappears in disguise.

Q9 — Buffon With a Long Needle

A needle of length $L = 2d$ is dropped on a floor ruled with parallel lines spaced d apart. Find (a) the expected number of lines it crosses and (b) $\mathbb{P}$(it crosses at least one line).

Solution. (a) *Linearity:* cut the needle into collinear pieces; the crossing count is the sum of the pieces' counts, so $\mathbb{E}[\#] = cL$ for a constant c. For a short needle ($L \leq d$) the count is 0 or 1, so $\mathbb{E}[\#] = \mathbb{P}(\text{cross}) = \frac{2L}{\pi d}$, which fixes $c = \frac{2}{\pi d}$. Hence for any L:

$$\mathbb{E}[\#\text{crossings}] = \frac{2L}{\pi d} = \boxed{\frac{4}{\pi} \approx 1.273}$$

(b) Let $\theta \in (0, \frac{\pi}{2})$ be the acute angle with the lines. The needle's vertical extent is $h = L\sin\theta = 2d\sin\theta$.

- If $h \geq d$ the needle must cross a line (probability 1).
- If $h < d$ it crosses with probability h/d (a segment of length h meets a line of spacing d).

The threshold is $\sin\theta_0 = \frac{1}{2}$, i.e. $\theta_0 = \frac{\pi}{6}$:

$$\mathbb{P} = \frac{2}{\pi}\Big[\int_0^{\pi/6} 2\sin\theta\,d\theta + \Big(\tfrac{\pi}{2} - \tfrac{\pi}{6}\Big)\Big]$$

$$= \frac{2}{\pi}\Big[2 - \sqrt{3} + \tfrac{\pi}{3}\Big] = \boxed{\tfrac{2}{3} + \tfrac{4-2\sqrt{3}}{\pi} \approx 0.837}$$

▶ *Speed trick:* $\mathbb{E}[\#]$ is always $\frac{2L}{\pi d}$, even for long needles. Only $\mathbb{P}(\geq 1)$ needs the case split, and $\mathbb{E} \geq \mathbb{P}(\geq 1)$ is not violated ($1.273 \geq 0.837$).

Q10 — Capstone: Estimating π by Random Darts

(X, Y) is uniform on the unit square and $R = \sqrt{X^2+Y^2}$. Find (a) $\mathbb{P}(R \leq 1)$, (b) $\mathbb{E}[R \mid R \leq 1]$, (c) $\mathbb{E}[X \mid R \leq 1]$, and (d) how many darts are needed so that the estimator $\hat{\pi} = 4\cdot$(fraction with $R \leq 1$) has standard error $\leq 10^{-3}$.

Solution. (a) The event is a quarter disk: $\mathbb{P} = \boxed{\frac{\pi}{4}}$.

(b) Given the dart is inside, $\mathbb{P}(R \leq r \mid R \leq 1) = \frac{\pi r^2/4}{\pi/4} = r^2$, so $f_R(r) = 2r$ and

$$\mathbb{E}[R \mid R \leq 1] = \int_0^1 2r^2\,dr = \boxed{\frac{2}{3}}$$

(c) In polar coordinates over the quarter disk, $\iint x\,dA = \int_0^{\pi/2}\cos\theta\,d\theta\int_0^1 r^2\,dr = \frac{1}{3}$, and dividing by the area $\frac{\pi}{4}$:

$$\mathbb{E}[X \mid R \leq 1] = \boxed{\frac{4}{3\pi} \approx 0.424}$$

(d) Each dart is a Bernoulli(p) with $p = \frac{\pi}{4}$, and $\hat{\pi} = 4\hat{p}$:

$$\text{Var}(\hat{\pi}) = \frac{16\,p(1-p)}{n} = \frac{\pi(4-\pi)}{n} \approx \frac{2.697}{n}$$

$$\text{SE} \leq 10^{-3} \iff n \geq 2.697\times 10^6 \;\Rightarrow\; \boxed{n \approx 2.7\times 10^6}$$

⋆ *Full-circle intuition:* the same area-ratio idea that computes probabilities also *prices the cost of simulating them.* Error shrinks only like $1/\sqrt{n}$: one extra decimal digit of π costs 100× more darts.

Drill order for review: redo Q1 and Q3 (curved boundaries) closed-book, then Q6 (state in one sentence why a fixed point lands in a bigger-than-average piece) and Q8 (the circle-to-stick reduction). Together they test whether area ratios, curved-boundary integrals and the stick-breaking simplex work as one connected skill, not four separate tricks.